

SAWARNEE SETHI

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Solutions Architect with 7+ years of experience at Amazon Web Services designing, deploying, and managing complex cloud solutions. Proven track record of translating business requirements into scalable technical architectures, driving multi-million dollar deals, and delivering measurable cost optimization for enterprise and public sector clients. AWS Direct Connect SME with deep expertise in networking, infrastructure, serverless, and AI/ML architectures.

CORE COMPETENCIES / KEY SKILLS

- **Certifications:** AWS Certified Solutions Architect - Professional & Associate | AWS Certified Advanced Networking - Speciality | AWS Certified AI Practitioner | AWS Certified Speaker Associate
- **AWS Services:** EC2, VPC, S3, RDS, IAM, CloudFormation, ECS, EKS, SES, Direct Connect, VPN, Transit Gateway, Network Firewall, Gateway Load Balancer, Route 53, Amazon Bedrock
- **SME Accreditation:** AWS Direct Connect
- **Architecture Domains:** Serverless, Microservices, Security & Compliance, Data Lake/Warehouse, Disaster Recovery, Analytics, Multi-Account/Multi-Region Design, Cost Optimization
- **Technical Domains:** Network Architecture (BGP/Static Routing, SD-WAN, DX/TGW), Cost Modelling & TCO Analysis, Infrastructure as Code, Well-Architected Reviews

PROFESSIONAL EXPERIENCE

DEC 2024 – PRESENT

SOLUTIONS ARCHITECT, AMAZON WEB SERVICES (AWS)

- **UNICEF - RapidPro Platform Automation:** Designed and deployed a scalable RapidPro infrastructure on AWS for UNICEF India, targeting rapid adoption by multiple government ministries. Developed a fully automated, single-click deployment using AWS CloudFormation for swift, repeatable provisioning of secure environments. Reduced implementation time and secured a strong technical relationship, positioning AWS for a 20x expansion of monthly billing pipeline.
- **ISHA Foundation - High-Scale Event and Application Launch Support:** Architected and validated the scaling strategy across ECS clusters, EC2, and Load Balancers for the ISHA's Mahashivaratri 2025 event, successfully handling a massive traffic surge that resulted in **1.4 million mobile application downloads in 12 hours with zero reported downtime**. Additionally, provided emergency architectural support to orchestrate a critical, last-minute promotional campaign targeting **7 million users** by securing and configuring a temporary increase in **SES sending rates to 500 TPS** over a 4-hour window, ensuring successful delivery of mission-critical marketing goals during the peak event.
- **Government Healthcare Agency - Strategic Cost Optimization Analysis:** Conducted a comprehensive Bill of Quantities (BOQ) analysis and technical validation for a major Government Healthcare Agency's critical workload renewal, benchmarking AWS against two major competing Cloud Service Providers (CSPs). Developed a tailored, deep-dive cost-optimization strategy that demonstrated an approximate **80% reduction** in total projected running cost compared to other major CSPs, effectively positioning AWS as the clear cost-to-performance leader in the competitive RFP landscape and directly contributing to a high-value government sales enablement motion.
- **Coal India Limited (CIL) - SAP RISE Migration Network Architecture:** Led the complete network architecture design and deployment for the strategic SAP ECC to RISE on AWS migration, a \$10M (6-year TCO) win. Engineered the high-throughput migration path by implementing **AWS Site-to-Site VPN** to facilitate the massive one-time data transfer. Subsequently, ensured enterprise-grade, low-latency, and highly available connectivity for **SAP users** by provisioning a resilient network architecture leveraging **multiple dedicated AWS Direct Connect links**. This foundational networking work was essential to securing the deal and unlocking a further \$18M in total cloud opportunity.
- **Echo India - Cost-Optimized Multi-Agent RAG Solution:** In collaboration with the Cloud Innovation Center team, designed and delivered a cost-optimized, multi-agent RAG chatbot utilizing **Strands SDK** and **Amazon Bedrock**. Engineered the solution to leverage an **S3 vector store**, realizing a **90% cost saving** compared to traditional vector databases (under \$2,000/month at maximum capacity). The successful two-domain pilot achieved **over 90% accuracy**, leading to the customer's decision to expand to a full **73-disease implementation** by the end of 2025.

- **Ministry of Women & Child Development (MWCD) - AI-Powered Nutritional Analysis for Poshan Tracker:** Built and demonstrated a POC using Amazon Bedrock (Claude) multimodal vision models to analyze meal photographs from Anganwadi Centers, automatically identifying food items, estimating portions, and scoring nutritional compliance against ICDS norms. Fully serverless architecture (Bedrock, Lambda, S3, DynamoDB) ensures **zero infra management, pay-per-use cost efficiency, and elastic scalability to 12L+ daily images across 14L centers serving ~9Cr beneficiaries**. The successful POC opened discussions on broader AI-driven image analysis use cases, including AI-based child height/weight estimation from images — currently in active development.
- **Federal Accounts – Network setup expertise:** Have been involved in some of the largest public sector opportunities like NIC migration to AWS Outposts (Meghraj 2.0) and SAP RISE workload migration to AWS for India PSUs (GAIL, CIL and BL).
- **Summits and demo booths:** Represented AWS at multiple showcase opportunities - India Impact AI summit, UP AI conclave (got the opportunity to present in front of UP CM), New Delhi Symposium, NHA NHCX hackathon and Arohana.
- **Workshops and Immersion days:** Delivered multiple sessions to different customers including 2-day EKS enablement session to Dhanush Healthcare and a session on CDN and Edge services to NIC officers at IIT campus, Greater Noida.

JUL 2019 – NOV 2024

CLOUD SUPPORT ENGINEER, AMAZON WEB SERVICES (AWS)

- **Hire And Develop:** 50+ interviews taken; 30+ trainings delivered.
- **AWS Enabler session 2023:** Delivered 1.5-hour session on “Advanced Computer Networking” to final year college students; session attended by more than 3000 students.
- **Case Reviews:** 100+ support cases reviewed (quality, technical and SME work evidence reviews).
- **Infrastructure Event Management:**
 - **IPL 2020/2021 and ICC World Cup 2020** - 20 days of IEM event support for 5 different customers.
 - **Melbourne Cup 2022** – 1 day support for 8 different customers.
 - **Leidos migration** – 2 days of standby support for critical ABIS workloads migration to AWS gov cloud.
- **Architectural reviews:** Complete Direct Connect setup review, service and features explanation, best practices, limitations, performance tests, future expansion and helping with redundant setup creation on AWS console [**Pepsico, Paytm, Singapore Press Holdings Limited**]
- **Trainings:** Delivered session on “Best Practices for AWS Site to Site VPN configuration” along with practical demos and new feature introduction [**SAP**]. Delivered sessions on AWS Direct Connect, Site to Site VPN connections and Transit Gateway as part of 4-day Networking bootcamp for customer’s technical team (160+ participants) [**Accenture**].
- **Well-architected designs:** Complete network architecture design involving services like VPC peering, TGW, VGW, VPN, and DX. Compared different architectural patterns as per different use-case scenarios on the complexity, limitations, future expansion and pricing aspect [**Godigit, Freemium**]. Discussion over advanced networking, static/BGP routing in DX/TGW architecture, SD-WAN connectivity with multiple on-prem locations and multiple AWS regions [**Maxeon**].
- **Escalations:** Handled some of the biggest escalations and complex troubleshooting scenarios involving multiple stakeholders and weeks/months of troubleshooting and testing [**Firstsource, Philip Futures**]. Also handled several E2M and Dive and Save escalations for languished cases [**Valuefirst**].

PROJECTS

- **Zero-Trust Dynamic Access Control at Scale:**
Architected a scalable, multi-account zero-trust network access solution using fully managed AWS services (Client VPN, Transit Gateway, Network Firewall, Lambda, DynamoDB) to enforce per-user, per-port access control with automated session lifecycle and full audit trail. Delivered a highly available, event-driven design supporting 400 users across 20 accounts at ~\$5/user/month — 50–75% cheaper than third-party ZTNA platforms — while eliminating operational overhead of managing per-account security rules.
- **Hub-and-Spoke Multi-Account Network Connectivity:**
Designed a highly available hub-and-spoke architecture enabling multi-account, multi-region workloads to route through a single central service VPC for internet egress, on-premises connectivity, shared VPC endpoints (SSM, S3, DDB), and Route 53 resolver access — consolidating networking resources and reducing per-account complexity.
- **Centralized Traffic Inspection for Egress and Ingress Workloads:**
Deployed a dual-layer inspection architecture: (1) AWS Network Firewall with Gateway Load Balancer for egress traffic intrusion prevention, identity-aware filtering, and TLS/HTTPS inspection; (2) Network Load Balancer with destination NAT through Checkpoint firewalls for deep packet inspection of ingress traffic to EKS workloads.

EDUCATION

B.Tech, Computer Science & Engineering | IPU Delhi | 2019 | **85.6%**